12-12 第1篇

阅读理解（说明文） Why Do We Laugh When We Are Tickled?

① One of the small funny mysteries in a child's life is this: a five-year-old knows, ahead of time, that her father is going to tickle her under the arm — she may even beg him, laughing, not to do it. Yet the moment his fingers touch her side, she is absolutely helpless with laughter. She cannot stop. She does not find it really funny (the joke hasn't even been told), and yet her whole body behaves as if it has just heard the best joke in the world. Why? Tickling, which looks like a small ordinary human joke, turns out to be one of the most carefully studied small reflexes in neuroscience. Behind the laugh are **three surprising findings** that even many adults have not heard before.

② **Finding one: there are actually two kinds of tickling.** Scientists carefully separate them. **Knismesis** is the light, feather-like tickle — the kind that makes you want to brush away an invisible insect. It does not, usually, produce laughter; it produces small, careful alertness. **Gargalesis** is the deeper tickle — pressure on certain sensitive spots of the body, which produces the big, uncontrollable laughter. Humans, great apes, and — surprisingly — **rats** are the only animals we know of that show something like gargalesis. (Rats, when tickled gently on the belly by a kind researcher, produce ultrasonic "giggle" sounds at about **50 kilohertz**, clearly audible only with special equipment. Research by psychologist Jaak Panksepp at Washington State University, 1998—2012.)

③ **Finding two: you cannot tickle yourself.** Try it. Run your own fingers under your own arm, against your own side. Nothing happens. Your body stays calm. The reason: the **cerebellum** — a small structure at the back of your brain — **predicts in real time the sensory results of your own movements** and sends a signal ahead to "cancel" them in your conscious experience. When you lift your hand, the cerebellum tells the rest of the brain: *"Expect a touch under the arm in 0.2 seconds — it's just us, not a surprise."* A 1998 University College London study (using fMRI scans of 24 adults) showed the brain region that normally activates in response to tickle-sensation (the **somatosensory cortex**) activates **far less** when the person tickles herself than when another person tickles her, even at the same physical force. Tickling needs **surprise** to work — and your own body always knows what you are about to do.

④ **Finding three: the laugh is not really "fun"; it is a social signal.** Modern neuroscience has spent decades looking inside tickle-laughter, and the honest answer is: the laugh is **not** the same as the laugh we make at a funny joke. **Joke-laughter** and **tickle-laughter** activate different regions of the brain: joke-laughter mainly the **frontal cortex** (humour understanding), tickle-laughter mainly the **hypothalamus + anterior cingulate cortex** (a social bonding area). A 2014 study at Tübingen University, Germany, scanned 40

volunteers: the tickle-laugh overlapped 37% with pain regions in the brain — unlike the clean joke-laugh, which did not. In other words, **tickle-laughter is not a smaller cousin of joke-laughter**. It is an ancient, almost involuntary response that, some scientists suggest, evolved as a **kind of friendly "mock-surrender"** — a small signal used between close family members to say, *"you have physically overpowered me, but this is play; we are safe."*

⑤ **What makes a spot ticklish?** Research on 1,068 adults at the University of Bristol (2019) mapped the most ticklish spots. The top five, in order, were: **under the arms, ribs, belly, the soles of the feet, the back of the neck**. What links them? Each is an area that is **(a) sensitive (many nerve endings in a small skin area), (b) close to vital organs** (heart, lungs, stomach, spinal cord), and **(c) relatively hard to defend** in a fight. Evolutionary biologists suggest the reflex may have once protected us: a surprise touch on a vital area **alerts the body quickly, draws attention to the threat, and, in play, teaches children how to protect these spots while laughing it off**. Tickling between older and younger siblings, or parent and child, may, some researchers argue, be a playful version of "defence training."

⑥ **What about people who "hate" being tickled?** They are not imagining the discomfort. A small but real minority of humans — around **6% according to the 2019 Bristol study** — find even light tickling deeply unpleasant. Their tickle-laugh does not contain the usual friendly-bonding marker, and cortisol (a stress hormone) rises rather than falls. Respect people who do not like it. The laugh, in their case, really is a small signal of distress, not a hidden joy.

⑦ **Why does tickling lose its power as we grow older?** Adults are less easily tickled than children, and adults being tickled by strangers laugh less than adults being tickled by close family. Two reasons. First, the cerebellum becomes more skilled at anticipating: adults predict the touch pattern better, reducing the surprise. Second, adult tickle-laughter requires **psychological safety** — the same 2014 Tübingen study found that the laugh sharply reduced when the scanner-subject was in an unfamiliar or stressful room. The reflex is, in a deep sense, a **safety signal between people who already trust each other**. When safety is low, no laugh comes.

⑧ **Is tickling a "real" form of communication?** The answer seems to be yes, in a quietly surprising way. In households where parents tickle children in a safe, short, playful way (never too long, never after a "stop"), children show slightly higher warmth scores toward parents in adolescence — a weak but real effect in a 2023 Zhejiang University longitudinal study of 824 families over 9 years. Tickling, well-used, is a small piece of the long language of family warmth. Ill-used (too long, ignoring a "stop"), it becomes the opposite: a breach of trust.

⑨ **The quiet rule.** Stop when the person says stop. Look at their face; a stressed laugh is not the same as a happy laugh. A tickle should last **seconds, not minutes**. When the rule is kept, tickling is what it has been for five or six million years of human family life: a

small, ancient, nearly-wordless way of saying, *"We are safe; we are close; we play."* And the helpless giggling of a five-year-old with her father's fingers light under her arm is not only a laugh — it is **a tiny living piece of human biology**, already watching the adults who will love her, already learning that some of the best messages we give each other do not, in the end, need any words.

注释： knismesis 轻触痒觉；gargalesis 深触痒觉；cerebellum 小脑；hypothalamus 下丘脑；cortisol 皮质醇

根据短文内容，选择最佳答案。

1. According to Paragraph ②, what is the difference between "knismesis" and "gargalesis"?

A. Both are the same thing.
B. Knismesis is the light feather-like tickle that produces alertness, not laughter, while gargalesis is the deeper pressure-based tickle that produces big uncontrollable laughter; humans, great apes, and rats are the only animals known to show something like gargalesis, with Panksepp's rats giggling in ultrasonic sounds at about 50 kHz.
C. Knismesis only affects animals.
D. Only adults feel gargalesis.

2. Why can we not tickle ourselves?

A. Because we don't try hard enough.
B. Because the cerebellum predicts in real time the sensory results of our own movements and sends a signal ahead to cancel them in our conscious experience; a 1998 UCL fMRI study found the somatosensory cortex activates far less for self-tickling than for another-person tickling at the same physical force — tickling requires surprise.
C. Because our fingers are too weak.
D. Because self-tickling is forbidden.

3. What did the 2014 Tübingen study find about tickle-laughter vs. joke-laughter?

A. They are exactly the same.
B. Tickle-laughter mainly activates the hypothalamus + anterior cingulate cortex (a social bonding area) and overlaps 37% with pain regions; joke-laughter mainly activates the frontal cortex (humour understanding) and does not overlap with pain regions — so tickle-laughter is not a smaller cousin of joke-laughter but an ancient almost involuntary response that evolved as a kind of friendly "mock-surrender."
C. Joke-laughter overlaps more with pain.
D. Neither uses any brain area.

4. What did the 2019 University of Bristol study of 1,068 adults find about ticklish spots and about the minority who hate tickling?

A. Only feet are ticklish; 90% of humans hate tickling.
B. The top five ticklish spots in order were under the arms, ribs, belly, soles of the feet, and back of the neck — each sensitive, close to vital organs, and hard to defend; about 6% of humans find even light tickling deeply unpleasant, with cortisol rising rather than falling during their tickle-laugh.
C. Nobody is ticklish.
D. 95% love tickling.

5. What is the main idea of the final paragraph?

A. Tickling is an ugly leftover of evolution and should be forbidden.
B. Tickling, when the quiet rule of "stop when the person says stop, watch the face, keep it seconds not minutes" is kept, is what it has been for five or six million years of human family life: a small ancient nearly-wordless way of saying "we are safe; we are close; we play"; a five-year-old's helpless giggle under a father's fingers is a living piece of human biology showing that some of the best messages we give each other do not, in the end, need any words.
C. Tickling has no purpose at all.
D. Only parents may tickle children.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第②段两种痒
【解析】"羽毛 vs 按压 + 鼠 50 kHz"直引。A、C、D 均错。故选 B。

2.
【答案】B
【题型】细节理解题
【定位】第③段自挠不痒
【解析】"小脑 / UCL fMRI / 惊喜"直引。A、C、D 均错。故选 B。

3.
【答案】B
【题型】细节理解题
【定位】第④段图宾根
【解析】"37% 与痛区重叠"直引。A、C、D 均错。故选 B。

4.
【答案】B
【题型】细节理解题

【定位】第⑤⑥段
【解析】"五大点 / 6% 不喜欢"直引。A、C、D 均错。故选 B。

5.
【答案】B
【题型】主旨大意题
【定位】第⑨段结尾
【解析】"安全 / 亲近 / 我们在玩"主旨。A、C、D 均偏离。故选 B。